

ARADHANA SINGH

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PROFESSIONAL SUMMARY

Aspiring Data Analyst with a strong quantitative research background (M.Sc. Bioinformatics, GATE-BT 2025 qualified) and hands-on experience turning complex, large-scale datasets into actionable business insights. Proficient in SQL, Python, and Power BI, with a portfolio of end-to-end analytics projects spanning employee attrition, customer segmentation, and revenue analysis. Skilled in writing advanced SQL (CTEs, window functions), building interactive dashboards, and translating technical findings into clear business recommendations. Completed data analytics and visualization job simulations with Tata Consultancy Services and Deloitte Australia.

TECHNICAL SKILLS

- Programming & Query Languages:** SQL (PostgreSQL), Python (Pandas, NumPy)
- BI & Visualization:** Power BI, Tableau, Microsoft Excel
- Statistical Analysis:** Correlation Analysis, Hypothesis Testing, Descriptive Statistics, Regression
- Data Analysis Techniques:** Data Cleaning & ETL, KPI Development, Customer Segmentation (RFM), Attrition/Churn Analysis
- Tools & Platforms:** Git, VS Code, Linux/Bash, PostgreSQL

PROJECTS

HR Employee Attrition & Retention Analytics | PostgreSQL, Python, Power BI

- Built an end-to-end attrition analytics pipeline analyzing **1,470 employee records** to identify key drivers of turnover, using 12+ SQL queries (CTEs, window functions) and validating findings in Python (Pandas, Seaborn).
- Identified that employees working overtime churn at **nearly 3x the rate** of others (30.5% vs 10.4%), the strongest predictor of attrition in the dataset, through correlation and segmentation analysis.
- Delivered an interactive Power BI dashboard with **5 custom DAX measures** and dynamic filters, enabling real-time attrition tracking across department, tenure, and overtime status.
- Quantified estimated annual attrition cost at **\$6.8M**, providing a data-driven business case for retention investment.

End-to-End E-Commerce Customer RFM Segmentation | PostgreSQL, Power BI

- Segmented **4,338 customers** across **500K+ transactions** using Recency, Frequency, and Monetary (RFM) analysis, enabling targeted marketing strategy, via SQL aggregations, window functions, and CASE WHEN logic.
- Identified that **18.1% of customers ("Champions")** generate over **58% of total revenue** (\$5.2M of \$8.91M), creating a clear customer-prioritization framework.
- Flagged **35.9% of customers as "Lost/Inactive,"** directly informing a targeted win-back email campaign strategy.
- Built an executive Power BI dashboard translating technical segmentation into a business-ready decision-making tool.

CERTIFICATIONS & VIRTUAL EXPERIENCE- [View certificates](#)

- Tata Data Visualization: Empowering Business with Effective Insights — Job Simulation, Forage
- Deloitte Australia Data Analytics— Job Simulation, Forage
- Complete Data Analytics — LearnTube by CareerNinja
- SQL — LearnTube by CareerNinja
- Power BI — LearnTube by CareerNinja
- OCI 2025 certified Generative AI Professional — Oracle University
- The Joy of Computing Using Python — IIT Madras, NPTEL
- Introduction to Machine Learning — IIT Kharagpur, NPTEL

PROFESSIONAL EXPERIENCE

Computational Research Associate — Moleculyst

Oct 2025 – May 2026

- Optimized High-Performance Computing (HPC) workflows using the PBS job scheduler, **reducing simulation turnaround time by 25%** across 50+ computational runs.
- Created and curated structured datasets from simulation outputs for machine learning model training, applying data preprocessing and quality-control best practices.
- Resolved **10+ software dependency and configuration issues** while deploying analytical tools on a national supercomputing cluster, strengthening technical troubleshooting and Linux/Bash proficiency.

EDUCATION

M.Sc. Bioinformatics — Banaras Hindu University, Varanasi (GATE-BT 2025 Qualified)	2023 – 2025
B.Voc. Medical Laboratory Technology — Banaras Hindu University, Varanasi	2020 – 2023